

DB-Lab12 Submission

Name: Yash Raj

Roll Number: 24k-0737

Task01:

```
> use SchoolDB  
< switched to db SchoolDB
```

Task02:

```
> db.createCollection("Students")  
< { ok: 1 }  
> db.createCollection("Courses")  
< { ok: 1 }  
SchoolDB>
```

Task03:

```
> db.Students.insertMany([ { _id: 1,  
  name: "Alice",  
  age: 20,  
  scores: {  
    math: 85,  
    science: 90 } }, { _id: 2,  
  name: "Bob",  
  age: 22,  
  scores: {  
    math: 78,  
    science: 82 } }, { _id: 3,  
  name: "Charlie",  
  age: 21,  
  scores: {  
    math: 92,  
    science: 88 } }, { _id: 4,  
  name: "Daisy",  
  age: 23,  
  scores: {  
    math: 68,  
    science: 74 } } ] )
```

Output:

```
< {
  acknowledged: true,
  insertedIds: {
    '0': 1,
    '1': 2,
    '2': 3,
    '3': 4
  }
}
```

Task04:

```
> db.courses.insertMany([
  {
    _id: 101,
    courseName: "Mathematics",
    instructor: "Dr. Smith",
    studentsEnrolled: [1, 2, 3]
  },
  {
    _id: 102,
    courseName: "Science",
    instructor: "Dr. Adams",
    studentsEnrolled: [2, 3, 4]
  }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': 101,
    '1': 102
  }
}
```

Task05:

```
> db.Students.find({$and: [{age: {$lt: 22}}, {"scores.math": {$gt: 85}}]})
< {
  _id: 3,
  name: 'Charlie',
  age: 21,
  scores: {
    math: 92,
    science: 88
  }
}
```

```
> db.courses.find({$and: [{instructor: "Dr. Adams"}, {studentsEnrolled: 3}]})
< {
  _id: 102,
  courseName: 'Science',
  instructor: 'Dr. Adams',
  studentsEnrolled: [
    2,
    3,
    4
  ]
}
```

Task06:

```
> db.Students.find({$and: [{"scores.math": {$gt: 80}}, {"scores.science": {$lt: 90}}]})
< {
  _id: 3,
  name: 'Charlie',
  age: 21,
  scores: {
    math: 92,
    science: 88
  }
}
```

```
> db.Students.find({$or: [{age: {$lt:23}}, {"scores.math":{$gte:85}}]})
< {
  _id: 1,
  name: 'Alice',
  age: 20,
  scores: {
    math: 85,
    science: 90
  }
}
{
  _id: 2,
  name: 'Bob',
  age: 22,
  scores: {
    math: 78,
    science: 82
  }
}
```

```
{
  _id: 3,
  name: 'Charlie',
  age: 21,
  scores: {
    math: 92,
    science: 88
  }
}
```

SchoolDB > |

```
> db.Students.find({$and: [{"scores.science":{$gte: 80}},{$or:[{"scores.math":{$lt: 80}}]})
< {
  _id: 2,
  name: 'Bob',
  age: 22,
  scores: {
    math: 78,
    science: 82
  }
}
```

Task07:

```
> db.Students.updateOne({name: "Bob"},{$set:{"scores.math":{$gte:75}}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Task08:

```
> db.Students.updateMany({"scores.science":{$lt: 80},age:{$gt: 22}},{$inc: {"scores.math": 5}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Task09:

```
> db.Students.deleteOne({name:"Daisy","scores.science":{$lt:80}})
< {
  acknowledged: true,
  deletedCount: 1
}
SchoolDB> |
```

Task10:

```
> db.Students.deleteMany({$or:[{studentsEnrolled:2},{instructor:"Dr.Smith"}]})
< {
  acknowledged: true,
  deletedCount: 0
}
```

Task11:

```
> db.Students.drop()
< true
SchoolDB> |
```

Task12:

```
> db.courses.drop()
< true
SchoolDB> |
```

Task13:

```
> db.dropDatabase()
< { ok: 1, dropped: 'SchoolDB' }
SchoolDB>
```